

Midterm Review

EC 441 — Lectures 1–8

EC 441 - Introduction to Computer Networking
Boston University

Spring 2026

Topics Covered

- 1 Network fundamentals & delays
- 2 Information theory & entropy
- 3 Physical layer: guided media
- 4 Digital signaling & bit error rate
- 5 Wireless physical layer
- 6 Error control coding
- 7 Cyclic Redundancy Check (CRC)
- 8 Multiple access protocols (MAC)
- 9 Ethernet (IEEE 802.3)

Exam format

Closed book. One instructor-provided reference sheet.

Topic 1: Network Fundamentals & Delays

Packet switching: data divided into packets; each routed independently (store-and-forward).

Four sources of delay:

Delay type	Formula	Typical scale
Transmission	$d_{trans} = L/R$	ms (links)
Propagation	$d_{prop} = d/v$	μs –ms
Queueing	depends on traffic	variable
Processing	inspection, forwarding	μs

L = packet size (bits), R = link rate (b/s), d = distance, $v \approx 2 \times 10^8$ m/s

Protocol layering (top to bottom): Application — Transport — Network — Link — Physical

Quick check

$L = 1500$ bytes, $R = 100$ Mb/s, $d = 2000$ km.

$d_{trans} = 12000/10^8 = 0.12$ ms. $d_{prop} = 2 \times 10^6/2 \times 10^8 = 10$ ms.

Topic 2: Information Theory & Entropy

Self-information: $I(p) = -\log_2 p$ bits (rare events = more information)

Shannon entropy:

$$H(X) = -\sum_i p_i \log_2 p_i \quad \text{bits/symbol}$$

Maximised when symbols are *equally probable*: $H_{\max} = \log_2 M$ for M symbols.

Data rates (uncompressed)

- Voice (telephone): 64 kb/s
- CD audio: 1.41 Mb/s
- 1080p video @ 30 fps: ~186 Mb/s raw;
3–25 Mb/s compressed

Prefixes

SI (decimal)	Binary
1 KB = 10^3 B	1 KiB = 2^{10} B
1 MB = 10^6 B	1 MiB = 2^{20} B
1 GB = 10^9 B	1 GiB = 2^{30} B

Topic 3: Physical Layer — Guided Media

Media comparison:

Medium	Bandwidth	Range	Use
Twisted pair	1–1000 MHz	~100 m	Ethernet, phone
Coaxial	1–1000 MHz	~500 m	Cable TV, legacy
Optical fiber	>10 THz	100+ km	Backbone

Attenuation: $\text{dB} = 10 \log_{10}(P_{\text{out}}/P_{\text{in}})$ (negative = loss; $-3 \text{ dB} \Rightarrow$ half power)

Shannon-Hartley capacity:

$$C = B \log_2(1 + \text{SNR}) \quad \text{bits/s}$$

B = bandwidth (Hz), SNR = signal-to-noise ratio (linear).

Quick check

$B = 100 \text{ MHz}$, $\text{SNR} = 63$ ($\approx 18 \text{ dB}$). $C = 10^8 \times \log_2(64) = 600 \text{ Mb/s}$.

Topic 4: Digital Signaling & Bit Error Rate

Line codes: NRZ-polar (balanced), Manchester (mid-bit transition, self-clocking, $2\times$ bandwidth).

M-ary signalling: M amplitude levels $\Rightarrow \log_2 M$ bits/symbol.

- More levels $\Rightarrow$ higher spectral efficiency but higher SNR required.
- Doubling M costs ≈ 6 dB in E_b/N_0 for same BER.

Matched-filter receiver (optimal for AWGN):

$$P_b = Q\left(\sqrt{\frac{2E_b}{N_0}}\right), \quad Q(x) = \frac{1}{\sqrt{2\pi}} \int_x^\infty e^{-u^2/2} du$$

“Waterfall” curve: BER falls steeply with increasing E_b/N_0 .

AWGN channel: $y(t) = s(t) + n(t)$; $n(t) \sim \mathcal{N}(0, \sigma^2)$, white, Gaussian.

Topic 5: Wireless Physical Layer

Free-space path loss (Friis):

$$P_r = P_t G_t G_r \left(\frac{\lambda}{4\pi d} \right)^2 \Rightarrow L_{FSPL}[\text{dB}] = 32.45 + 20 \log_{10}(f_{\text{MHz}}) + 20 \log_{10}(d_{\text{km}})$$

Link budget: $P_r[\text{dBm}] = P_t[\text{dBm}] + G_t[\text{dBi}] + G_r[\text{dBi}] - L_{FSPL}[\text{dB}] - L_{\text{misc}}$

QAM constellations: symbols in I-Q plane. BPSK (1 b/sym) $\rightarrow$ QPSK (2 b/sym) $\rightarrow$ 16-QAM (4 b/sym) $\rightarrow$ 64-QAM (6 b/sym).

Cellular frequency reuse:

$$D = R\sqrt{3N}, \quad \frac{C}{I} \approx \frac{(3N)^{n/2}}{6}$$

N = reuse factor, R = cell radius, n = path-loss exponent. Classic: $N = 7$. Modern systems: $N = 1$ with interference management.

Topic 6: Error Control Coding

Block code (n, k) : k data bits $\rightarrow$ n -bit codeword; code rate $R_c = k/n$.

Hamming distance: number of bit positions where two codewords differ.

Minimum distance d_{min} : smallest distance between any two valid codewords.

d_{min} **theorem**: to detect e_d errors *and* correct e_c errors simultaneously:

$$e_c + e_d \leq d_{min} - 1, \quad e_c \leq e_d$$

Goal	Required d_{min}	Example
Detect d errors only	$d + 1$	Parity: $d_{min} = 2$, detects 1
Correct d errors only	$2d + 1$	Hamming: $d_{min} = 3$, corrects 1

Detection vs. correction: detection + ARQ $\approx 5\times$ more efficient than FEC for typical network BERs.

Topic 7: Cyclic Redundancy Check (CRC)

Polynomial representation: bit string $\leftrightarrow$ polynomial over $\text{GF}(2)$.

Arithmetic is *modulo-2*: addition = XOR, no borrows/carries.

Encoding (degree- r generator $G(x)$):

- ① Shift message left r bits: $x^r M(x)$
- ② Divide by $G(x)$ (modulo-2): obtain remainder $R(x)$
- ③ Transmit $T(x) = x^r M(x) + R(x)$ (divisible by $G(x)$)

Decoding: divide received word by $G(x)$; remainder $\neq 0 \Rightarrow$ error detected.

CRC-32 used by Ethernet, WiFi, ZIP.

Error detection guarantees (with suitable $G(x)$):

- All single, double, odd-number bit errors
- All burst errors of length $\leq r$
- Most burst errors of length $> r$ (probability $1 - 2^{-r}$ or $1 - 2^{-(r-1)}$)

Topic 8: Multiple Access Protocols (MAC)

ALOHA throughput (G = offered load in frames/slot):

$$S_{pure} = Ge^{-2G} \quad (\text{max } 18.4\% \text{ at } G = 0.5), \quad S_{slotted} = Ge^{-G} \quad (\text{max } 36.8\% \text{ at } G = 1)$$

CSMA/CD (classic Ethernet):

- 1 Sense channel; transmit if idle.
- 2 Detect collision while transmitting $\Rightarrow$ jam (48 bits), abort.
- 3 **Binary exponential backoff**: after k -th collision, wait $r \cdot T_{slot}$ where $r \in [0, 2^{\min(k,10)} - 1]$.

Minimum frame size: must still be transmitting when collision echo returns.

$$L_{min} = 2\tau R \quad \Rightarrow \quad 64 \text{ bytes at } 10 \text{ Mb/s over } 2500 \text{ m}$$

Modern Ethernet: full-duplex switched — CSMA/CD not used in practice.

Topic 9: Ethernet (IEEE 802.3)

Frame format:

Preamble FCS 7 B 4 B	SFD 1 B	Dst MAC 6 B	Src MAC 6 B	Type/Len 2 B	Payload 46--1500 B
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MAC addresses: 48-bit, written hex (e.g. A4:C3:F0:85:AC:2D).

OUI = first 3 bytes (manufacturer). Broadcast = FF:FF:FF:FF:FF:FF.

ARP: broadcast “Who has IP x?” → unicast reply with MAC. Cached (~20 min TTL).

Switch self-learning:

- Record *source* MAC → ingress port on every frame.
- Unknown destination ⇒ **flood** all other ports.
- Known destination ⇒ **forward** to correct port only.

Speeds: 10 Mb/s (Manchester) → 100 Mb/s (4B5B+MLT-3) → 1 Gb/s (4D-PAM5) → 10/100/400 Gb/s.

Exam: Thursday during class

Closed book. Instructor-provided reference sheet.

You may bring a calculator.